

ITIS 3140 Midterm Study Guide

Exam: Thurs Oct 1, in-class, closed book, Lockdown Browser, 14 questions / 40 pts / 45 min — MC, fill-in-the-blank, matching only (no open-ended). **Quant Data is worth more than half the points**, so that section goes deepest.

1. QUANTITATIVE DATA (highest weight — know this cold)

Types of Measurement (memorize this hierarchy)

- **Categorical**
 - **Nominal** — named categories, no order (e.g., blood type: A/B/AB/O)
 - **Ordinal** — ranked categories, but gaps between ranks aren't equal (e.g., SEQ rating 1–7)
 - **Dichotomy** — special case of categorical with only 2 possible values
- **Numerical**
 - **Interval** — numeric, equal spacing, no true zero (e.g., Year: 2000, 2026)
 - **Ratio** — numeric, equal spacing, **has a true zero** (e.g., Height)

Quick check: "Blood type?" = Nominal. "SEQ?" = Ordinal. "Year?" = Interval. "Height?" = Ratio.

Types of Variables

- **Discrete** — distinct, separate values (countable)
- **Continuous** — any value within an interval (measurable)

Descriptive Statistics ("describing the data you have")

- Totals, min/max, **mean, median, mode, range**
- Example phrasing you might see: "mean age 22.4, range 18–37" or "median time 34.5s (min=19.2, max=305)"

Averages — which one to use with which data type

Measure	Best used with	Notes
Mean	Interval or Ratio	Sensitive to outliers
Median	Ordinal, Interval, or Ratio	Middle value when ranked; less sensitive to outliers
Mode	All types (Nominal too)	Most frequent value; can have more than one

Designing an Experiment — 4 things to define

1. What are your variables?
2. What is your experiment design?
3. What are you measuring?
4. Who are your participants?

Independent vs. Dependent Variables

- **Independent Variable (IV):** the thing you manipulate (presumed **cause**) — e.g., Device Type (Laptop vs. Phone)
- **Dependent Variable (DV):** the thing you measure (presumed **effect**) — e.g., # of Errors while typing
- Hypothesis example: "People commit fewer errors typing on laptops than phones." → IV = device type, DV = # of errors.
- **Confounding/extraneous variable:** something else that could

bias the DV if not controlled (e.g., environment, keyboard type).
Must be kept constant or controlled.

Experimental Design: Within- vs. Between-Subjects

	Within-Subjects	Between-Subjects
Setup	Every participant experiences every condition	Different participants assigned to different conditions
Pros	Fewer participants needed; removes individual-difference noise; minimizes learning effect issues (if counterbalanced)	Shorter sessions per participant; no carryover/practice effects between conditions
Risk	Ordering/practice/fatigue effects	Needs more participants; individual differences add noise

Controlling Variables — 3 methods

1. **Don't allow it to vary** (e.g., only test iPhone users)
2. **Allow it to vary randomly** (random assignment to groups)
3. **Counterbalance** — systematically vary it (e.g., equal males/females per group)

Randomization

- Random assignment of participants to conditions (A/B/C/D groups) eliminates:
 - **Experimenter effect** (researcher bias/differences between researchers)
 - Some **participant effects**

Ordering Effects (within-subjects problem) &

Counterbalancing

- Doing conditions in the same order every time → biased data from:
 - **Practice effect, fatigue**, other confounds (time of day, lighting, etc.)
- **Counterbalancing** = systematically alternate the order across participants (e.g., half do Phone→Laptop, half do Laptop→Phone). Assumes order effects are roughly symmetric/cancel out.

Web Experiments

- Large-scale experiment; another way to evaluate a live website/app
- Commonly used to evaluate **design changes** (e.g., A/B testing)
- Compares two (or more) groups of users on two (or more) design versions

Inferential Statistics

- Goal: make inferences about a **larger population** you didn't directly measure (vs. descriptive stats, which just describe your sample)
- Comparing two means: ask — given the variability in the data, how likely is it this difference happened **by chance**?

Hypothesis Testing & Significance

- **Significance level (p / "alpha")**: probability your result happened by chance even if the null hypothesis were true
 - Commonly set at **p = 0.05** (5%)
- **Which test to use (know these pairings — likely matching material)**:
 - **Chi-squared (χ^2)** → compare expected **frequencies**, used with **categorical** data
 - **t-test** → compare **two means**

- **ANOVA** (Analysis of Variance) → compare **several (3+) means**
- Don't assume normal distribution unless $n > 30$ (or you've run a normality test)
- "Paired/matched" means = within-subject design
- On the reference chart: columns = dependent variables, rows = independent variables

Practice scenarios from lecture (know how to map scenario → test):

- 15 users rate 2 app designs each (same people, both designs) on a 7-pt SEQ → **within-subjects, comparing 2 means → paired/matched t-test**
 - 30 students randomly split into 3 groups, 3 teaching methods, scores ~normal → **between-subjects, 3+ groups/means → ANOVA**
 - 50 voters surveyed Yes/No before and after a debate (same people, categorical, 2 time points) → **categorical/frequency comparison → Chi-squared**
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2. RESEARCH METHODOLOGY BASICS (Grad Coach video)

- **Qualitative** = non-numeric (words, ideas, perceptions); subjective; explores the "soft side"; often **inductive** (builds theory from data); used for **exploratory** aims
- **Quantitative** = numeric/statistical; objective; **deductive**; tests hypotheses; used for **confirmatory** aims
- **Mixed methods** = combines both — e.g., qualitative to build hypotheses, then quantitative to test them (or vice versa: quant for the "what," qual for the "why")
- **3 factors that determine which methodology to use:**
 1. Nature of your research aims/questions (exploratory vs. confirmatory vs. both)
 2. Disciplinary norms / what similar studies use

3. Practicalities & constraints (data access, time, money, equipment/software, your skills)
 - Rule of thumb: exploratory → qualitative; confirmatory → quantitative; both → mixed methods (but mixed is harder to execute well — a well-done mono-method beats a poorly-done mixed method)
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3. UX METHODS — "CHARACTERIZING" FRAMEWORK

Repeated for nearly every method in the slides — expect matching/fill-in questions using these 4 dimensions:

1. **Stage of Design:** Discover → Define → Develop → Deliver (the Double Diamond)
2. **Type of information:** Qualitative or Quantitative
3. **Activity type:** data gathering, analyzing, representing, or idea generating
4. **Who's involved:** Designers? Users? Experts?

Apply this lens to every method below — the exam likely tests whether you can classify each method along these 4 axes.

Method	Stage	Qual/Quant	Activity	Who
Mind Mapping	Discover/Define	Qualitative	Idea generating/representing	Designers (can include users)
Forced Association	Develop	Qualitative	Idea generating	Designers
Competitor Analysis	Discover	Both (reviews=qual, testing=quant possible)	Data gathering + analyzing	Experts/Designers

Cognitive Walkthrough	Define/Develop	Qualitative, predictive	Analyzing	Experts (product, UX, domain)
Heuristic Evaluation	Define/Develop	Qualitative, predictive	Analyzing	
Usability Testing	Develop/Deliver	Both	Data gathering	Users + Moderator
Interviews	Discover	Qualitative (mostly)	Data gathering	Users/ Stakeholders
Focus Groups	Discover/Develop	Qualitative	Data gathering	Users/ Stakeholders

4. MIND MAPPING & FORCED ASSOCIATION

- **Mind Mapping — 5 rules:** central concept in middle; each addition anchored; use color/shapes for memorability; keywords not full sentences; all text aligned (no upside-down text). Digital or paper.
- **5W+1H technique:** Who / What / Where / When / Why / How is the product/service used
- **Forced Association:** novices fixate on first ideas → mediocre solutions. Technique: combine disparate/random concepts (often pulled from a mind map) to spark lateral thinking. Stems from **speculative design** — doesn't need to be practical, but must still contribute to the main design goal.
- Novice vs. Expert designers (Mathias, 1993): novices go through steps linearly; experts loop — they revisit **analysis of the problem statement** and cycle through **convergence/solution concept** (this maps to Mind Maps = analysis stage; Forced Association = convergence/solution concept stage).

5. COMPETITOR ANALYSIS

- Understand where your product fits vs. others in the market; identify patterns/standouts
- Can be holistic or feature-specific
- "Competition" must be defined **broadly** (e.g., Uber's competitors = Lyft, public transit, Lime/Spin scooters, walkable areas)
- **Representations:**
 - **SWOT** (Strengths/Weaknesses = internal; Opportunities/Threats = external) — for a **single** competitor
 - **Table format** — for **multiple** competitors side by side
- **Two data-gathering methods:**
 - **Competitive Reviews** = expert review; less logistics, but biased/subjective
 - **Competitive User Research** = actual usability testing of competitors; more objective, more time-intensive
- **Imitation trap:** don't just copy competitors — look for gaps/opportunities beyond what they offer.

6. UX MEASURES

- **3 categories of usability measures:**
 1. **Performance-based** (objective): task success, time on task, error rate, error recovery rate, efficiency, learnability, # of navigations to help/manual
 2. **Issue-based:** # and type of issues, severity, frequency of encountering an issue
 3. **Self-reported perceptions:** SUS, CSAT, Frustration score, SEQ (Single Ease Question), NASA-TLX
- **SUS (System Usability Scale):** 10 questions, alternating negative/positive statements (Q1 positive, Q2 negative, Q3 positive...) — this alternation reduces response bias / forces the respondent to actually read each item rather than straight-lining answers.
 - Score bands (0–100): below ~50 = not acceptable/F; ~50–70 = marginal (D/C); 70+ = acceptable (B/A); scores near 100 = best imaginable.
- **Eye tracking metrics:** dwell time, number of fixations, fixation duration, sequence, time to first fixation, hit ratio — used to build

heat maps.

7. DISCOUNT EVALUATION: HEURISTIC EVALUATION & COGNITIVE WALKTHROUGH

- **Discount evaluation** = predicting usability instead of directly observing users (quick, low-cost, uses HCI experts). Works best when experts haven't used the prototype before, know the domain, and understand real users.
- **Heuristic Evaluation process:** design team provides prototype + set of heuristics → experts individually step through and log problems → design team compiles a master list + severity ratings → team decides on changes.
- **Nielsen's 10 Heuristics** (know these — likely matching!):
 1. Visibility of system status
 2. Match between system and the real world
 3. User control and freedom
 4. Consistency and standards
 5. Error prevention
 6. Recognition rather than recall
 7. Flexibility and efficiency of use
 8. Aesthetic and minimalist design
 9. Help users recognize, diagnose, and recover from errors
 10. Help and documentation (Note: slide order differs slightly from the "official" Nielsen list — study the *meaning* of each, not just the number, since exam order may vary.)
 - Example mappings from class: no loading spinner after tapping "Order Now" → **visibility of system status**; can't delete one cart item, must restart whole order → **user control and freedom**; system accepts a letter in a phone-number field → **error prevention**.
- **Cognitive Walkthrough (CW):** assesses **learnability** by simulating how a *new* user would explore an interface. Qualitative, predictive, often done in groups/workshops with experts (product experts, UX designers, engineers, domain experts).

- **Requirements:** description of users/background, description of the task, complete list of actions needed, prototype or system description.
- **The 4 CW questions**, asked at each step of a task:
 1. Will the user be trying to achieve the right result/effect?
 2. Will the user notice the correct action is available (is it visible)?
 3. Will the user associate the correct action with the result they want?
 4. Will the user understand the feedback after taking the action?

8. USABILITY TESTING

- ISO definition: usability = extent a product can be used by specified users to achieve goals with **effectiveness, efficiency, and satisfaction** in context.
- **Why test instead of just asking users?** People's stated behavior ≠ actual behavior — they don't notice their own struggles, rationalize after the fact, and can't always explain why something is hard.
- **Summative vs. Formative evaluation:**
 - **Summative** = after design is complete; assess whether goals were met, find areas to improve
 - **Formative** = during design; gather data on users/tasks, get feedback on concepts
- **Types of usability testing:** in-person moderated (lab or user's environment), remote moderated, remote unmoderated.
- **How many participants?** No fixed number — depends on qualitative vs. quantitative goals, schedule, participant availability, cost. Rule of thumb: **5–10 per segment**; some argue test until no new insights emerge.
- **Test script structure:** intro/procedures → pre-task questions → set of tasks → additional measures/insights → thanks.
- **Verbal protocols (know these!):**
 - **Think-aloud** — participant voices thoughts live. Adv: real-

- time insight into thinking. Disadv: interrupts the task.
- **Cooperative approach** — two participants talk to each other while doing the task (peers or coach+novice). More natural, but must fit the task.
- **Post-hoc/Retrospective think-aloud** — review a video recording after. Adv: doesn't interrupt task. Disadv: subject to memory/reinterpretation. Variant: **critical incident logging** (logging positive/negative moments).
- **Structured interview** — good open-ended questions after ("What did you like best/least?", "How would you change...?")
- **Validity types (likely matching):**
 - **Internal validity** — does the study correctly establish cause-and-effect?
 - **External validity** — do findings generalize beyond this study?
 - **Ecological validity** — do the study environment/tasks reflect real-world use?
- **Informed consent** — tells participants why the study is happening, what they'll do, and their rights; typically approved by an **IRB** (Institutional Review Board).

9. REPORTING USABILITY STUDIES — the 6 steps

1. **Data Tabulation** — put all measures into a table
2. **Analyze data** — descriptive stats (totals, means, medians) + qualitative thematic analysis (patterns of problems/confusion)
3. **Determine findings** — group evidence into findings that answer your original goals; prioritize by frequency/impact/ease-of-fix ("quick wins")
4. **Make recommendations** — depends on design stage (competitive analysis → what features/gaps to address; evaluation of existing design → root causes + what to improve)
5. **Make the report** — clear objectives/methods, summary of measures, findings, recommendations, visuals
6. **Archive the findings** — keep data for future benchmarking/

comparison (subject to IRB storage rules)

10. INTERVIEWS & FOCUS GROUPS

- **Interviews general tradeoffs:** rich detail, flexible questioning, but time-consuming to run/analyze, requires interpretation, subject to bias (avoid "cherry-picking" quotes/data).
- **Structured** — like a questionnaire, same set questions; needs lots of prep but gives **repeatable** results (inter-interviewer reliability).
- **Unstructured** — plan but no fixed questions; open-ended, participant-led, uses their own terms; time-consuming but rich.
- **Semi-structured** — guiding questions planned in advance (how/why/under what conditions/what info do you need/who do you talk to/what happens after...) — most common in practice.
- **User Interview vs. Usability Test (know this comparison table!):**

	User Interview	Usability Test
Purpose	Generate new knowledge about users	Assess a design
Research type	Attitudinal (reported thoughts/feelings)	Behavioral (observed interaction)
When used	Empathize/discovery stage	Test stage
Design present?	Absent	Present

- **Challenges to interviews:** building rapport; asking good probing/follow-up questions; not multitasking (record, don't just take notes); avoiding leading/priming questions (stay open-ended, watch body language).
- **Props** — devices to prompt the interviewee (e.g., a prototype or scenario card).
- **Stakeholder interviews** (industry use): gather context/history,

identify business goals & success metrics, build buy-in/ communication — same structure as user interviews.

- **Focus Groups:** getting participants to bounce ideas/reactions **off each other** (not just off a moderator); typically 3–10 people; can include users and stakeholders.
 - **Facilitating focus groups:** build rapport (warm-up activity), be an active listener, balance letting people talk vs. moving on, redirect circular discussion, draw out quiet members, consider adding an activity (not just talk) to boost engagement, make sure recording captures everyone.
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Fast self-check (cover the right column and quiz yourself)

- Nominal vs. Ordinal vs. Interval vs. Ratio → example for each
- IV vs. DV vs. Confounding variable
- Within-subjects vs. Between-subjects — pros/cons of each
- Chi-squared vs. t-test vs. ANOVA — when to use each
- $p = 0.05$ — what does it mean?
- Mean vs. Median vs. Mode — when to use each
- 4 Cognitive Walkthrough questions, in order
- Nielsen's 10 heuristics — be able to match a scenario to a heuristic
- SUS — why alternating pos/neg questions
- Think-aloud vs. Retrospective think-aloud vs. Cooperative approach
- Formative vs. Summative evaluation
- Internal vs. External vs. Ecological validity
- User Interview vs. Usability Test — purpose/research type/when used/design present
- The 4-question "Characterizing UX Methods" framework applied to each method covered

Good luck — given the weighting, if you only have time for one deep review pass, make it the **Quant Data** section above.